

NurseryConnect

Spatial

visionOS — Situational Awareness for Nursery Keyworkers

Multi-Window

Volumetric Charts

Real-Time Data

Spatial UX

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Concept Overview

What is NurseryConnect Spatial?

NurseryConnect Spatial is a purpose-built visionOS app that gives nursery keyworkers a hands-free, always-visible operational dashboard within Apple Vision Pro. While the iPad app provides a deep multi-panel workspace for focused data entry, the spatial experience is optimized for ambient situational awareness, understanding the whole nursery at a glance without picking up a device.

Shared Domain Layer

Reuses Core Data entities, GDPR-scoped fetch requests, and analytics services from the iOS target. No duplicate business logic.

Multi-Window Model

Hub window opens Keyworker Dashboard and Setting Manager as independent spatial windows — each floats independently in the user's space.

Read-Optimized UX

Spatial target is deliberately read-only for most workflows. Diary entry, full incident creation, and background sync are not included — spatial is for oversight, not data entry.

In-Memory Demo Store

Uses seedSpatialDemoStore with realistic wellbeing patterns for the volumetric mood chart. Production will use a shared CloudKit container.

Target Audience

Primary Personas

Nursery Keyworker

Front-line childcare practitioner managing 3–4 assigned children. Uses the app for real-time attendance status, diary completeness checks, and incident awareness without interrupting floor activity.

Setting Manager

Nursery manager responsible for whole-room oversight. Uses the Setting Manager window for cohort-level attendance, engagement metrics, dietary register, and keyworker performance monitoring.

Spatial-Specific Use Cases

1. Keyworker in nursery room checks at-a-glance which children are on site, awaiting, or absent — Vision Pro mounted on a workstation or worn passively
2. Setting Manager reviews parent engagement chart and open incident count across the whole room before a staff briefing
3. Quick "Open Mood Chart 3D" action surfaces a volumetric 7-day wellbeing chart for a specific child, rotatable with hand gestures
4. End-of-shift: keyworker taps End of Day Summary in the keyworker window without touching an iPad or phone

Spatial Computing Features

Multi-window Scene Model

Four independent WindowGroup scenes (Hub, Keyworker Dashboard, Setting Manager, Mood Chart). Keyworkers open only the windows relevant to their role. Each window uses .plain style + NCSpatialWindowChrome for consistent glass clipping.

WindowGroup, openWindow, dismissWindow

Volumetric Mood Chart (RealityKit)

Opens as a volumetric window ($0.48 \times 0.58 \times 0.42$ m) anchored in physical space. 3D bar chart shows 7-day wellbeing per child. Drag-to-rotate interaction. Welfare warning surface if avg mood < 2.5 .

RealityView, volumetric window group, drag gesture

visionOS Hover Effects

SpatialHubView experience cards use .hoverEffect() to provide tactile pre-selection feedback without requiring a tap — reducing accidental activations when keyworkers review data with a quick glance.

.hoverEffect(), .onHover()

Spatial Window Chrome

NCSpatialWindowChrome clips each window to a 28 pt corner radius using #if os(visionOS) conditional compilation, matching Apple's recommended ornament and glass background guidelines without affecting the iOS target.

#if os(visionOS), .clipShape(RoundedRectangle)

Read-Only Safety Constraint

visionOS target excludes sync services, incident composer, and diary editor — by design. This prevents accidental data mutation from a shared or workplace device. Production would add biometric auth for write operations.

Target membership exclusion, in-memory store

Shared Domain Layer

Core Data entities, analytics services, messaging view models, GDPR fetch predicates, and Swift Charts views are all compiled into the Spatial target unchanged — ensuring feature and data parity with the iOS app.

Shared source file target membership

Key Windows

spatial-hub

Spatial Hub

Nursery home screen. Greeting, today snapshot (children on-site, open incidents, diary gaps), and launch cards for each spatial experience.

~920 × 640 pt (plain window)

Pull-to-refresh via Setting Manager VM · visionOS hover cards · Nursery branding

keyworker-dashboard

Keyworker Dashboard

Operational view for the floor keyworker. Unread messages, parent engagement chart, cohort child picker with attendance/mood/diary status, quick actions, and recent incidents panel.

~980 × 720 pt (plain window)

Per-child aggregates (GDPR scoped) · Message thread sheet · Open Mood Chart 3D button

setting-manager

Setting Manager

Nursery-wide overview for room leaders. Total enrolled, attendance %, open incidents, 7-day engagement chart, keyworker cohort rows, dietary & allergy register.

~980 × 720 pt (plain window)

SpatialDietaryAlert badges · Nursery-wide incident review sheet · Cohort comparison

mood-chart

Volumetric Mood Chart

Child-specific 3D wellbeing visualisation. 7-day bar chart rendered in RealityKit on device (volumetric window); plain 2D fallback in Simulator. Drag-to-rotate, welfare threshold line.

0.48 × 0.58 × 0.42 m (volumetric)

RealityView 3D bars · Drag gesture rotate · Welfare warning < 2.5 avg

Volumetric Mood Chart & 3D Features

MoodChartVolumeView

- Volumetric window: $0.48 \times 0.58 \times 0.42$ m (device)
- Fallback: plain window on simulator (MoodChartBarStripView)
- 7-day wellbeing data from AnalyticsDataService + Core Data
- RealityView renders 3D bar columns per day — height mapped to mood score 1–5
- Welfare threshold plane (score < 2.5) with warning label
- Drag-to-rotate via DragGesture on RealityView content
- Opened per-child via `openWindow(MoodChartWindowID)`

Bundled 3D Assets (USDZ)

`child_avatar.usdz` — avatar placeholder under `NurseryConnectSpatial/Avatars/FoodModels/*.usdz` — plate and meal models for future diary integration
No Swift RealityKit references yet — staged for future USDZ integration sprint

Simulator vs Device

`#if targetEnvironment(simulator)` guard in `MoodChartVolumeView` ensures plain 2D chart renders when running on macOS (no RealityKit support). Device build uses `ModelEntity` + `RealityView` for true 3D volumetric bars. This conditional compilation pattern is required because RealityKit crashes in the visionOS simulator.

Design Decisions & Honest Constraints

Key Design Decisions

Separate @main target

NurseryConnectSpatialApp has its own entry point, excluding iPhone/iPad-only files (sync services, AdaptiveRootView) via Xcode target membership

Read-only spatial

Incident creation and diary entry are iPad/iPhone-only — spatial is for oversight. Protects data integrity and simplifies GDPR audit trail

Plain windows only

All spatial windows use .plain style + .ncSpatialWindowChrome() glass clipping rather than immersive spaces — appropriate for a work tool used alongside physical nursery activity

Demo in-memory store

seedSpatialDemoStore + applySpatialDemoSuccessState give a polished demo without persisting identifiable child records to CloudKit on shared devices

Known Constraints

CloudKit not wired

CloudKitSyncConfiguration exists with container ID + notification name. PersistenceController not yet updated to use. NSPersistentCloudKitContainer for spatial target, production task remain

USDZ assets unused

child_avatar.usdz and FoodModels/*.usdz are bundled but have no Swift RealityKit references — positioned for a future diary 3D meal-visualization sprint

Spatial target scope

SpatialIncidentsSheet is read-only; no full incident composer. A full spatial incident form would require body-map 3D interaction design beyond current scope

Simulator limitations

Volumetric mood chart requires a physical Vision Pro device; macOS simulator gets the 2D bar-strip fallback

Learning Reflection & Curated Resources

01

WWDC23 — Meet SwiftUI for spatial computing

↳ Introduced the SwiftUI APIs for visionOS — ornaments, .plain windows, hover effects, and Scene manifest. Directly informed how NurseryConnect Spatial lays out its hub cards and applies visionOS-specific modifiers behind #if os(visionOS) guards.

02

WWDC23 — Design for spatial user interfaces

↳ Covered the 7 spatial design principles: familiarity, wayfinding, scale, light, comfort, focus, and immersion. The decision to use .plain windows (not immersive spaces) and keep the hub compact (920×640 pt) came directly from the 'comfort' and 'wayfinding' principles.

03

WWDC23 — Develop your first immersive app

↳ Step-by-step RealityKit + RealityView walkthrough. The MoodChartVolumeView volumetric bar implementation — ModelEntity cylinders, AnchorEntity, and DragGesture rotation — follows the patterns from this session.

04

Apple Developer — Creating multi-window visionOS apps

↳ Multi-window openWindow / dismissWindow pattern with WindowGroup scene IDs. Informed the window ID enum (SpatialWindowID) and the hub's use of @Environment(\.openWindow) to launch Keyworker Dashboard and Setting Manager.